Week 86: Pursuit and escape

Adult guide built from the problem files · Unpiloted; counters, turn tiles and fold covers physically unrehearsed.

This view repeats, by problem, the answer keys of the print guide, week-86-facilitator.pdf, whose source is editable/guide/facilitator.tex; page references are to that guide. Change both together; check_guide_copy.py compares them.

Mathematical overview

The game uses a finite connected undirected graph. C chooses a starting dot, R chooses after seeing C, and C moves first. Turns alternate; each player may stay or cross one road. Positions are visible. Sharing a dot is immediate capture, including at placement. An evasion theorem means a response strategy forever, not survival of a classroom timer.

Every finite tree is cop-win. C moves toward R along their unique path. Delete C's current dot and look at R's component. Each uncaptured cop advance makes that component strictly smaller, and R cannot cross C to escape it. Finiteness gives capture. On an n-dot path the best worst-case start is a middle dot, with $\lfloor n/2 \rfloor$ cop moves: a farthest endpoint robber who stays gives the lower bound, and advancing from the middle gives the upper bound.

A bare cycle with at least four dots is robber-win. Before each cop turn R can remain outside C's closed neighborhood (C and its neighbors). After C moves, R stays if safe, or moves away to the other ring neighbor, which is safe. A triangle is different: one cop is already adjacent to every dot. The existence of a cycle inside a larger graph alone does not decide who wins.

A dominated dot may be folded away. If $N[v] \subseteq N[u]$ for distinct dots, deleting v preserves cop-win status; staying counts in these closed neighborhoods. A sequence of folds to one dot certifies a cop strategy. The full finite classification says a connected graph is cop-win exactly when it can be reduced this way. This deeper result is an adult destination, not a prerequisite for initial games. The explanation and limits appear with the fold solution.

One diagonal makes a square cop-win; a bare pentagon needs two added chords. One pentagon chord still leaves a square as a retract. Two cops catch on every finite cycle: one blocks a dot while the other sweeps the remaining path. These statements do not claim two cops suffice on every graph.

Problems and routes

Problem	2–5	4–5	route
Best cop start on two paths	P1	P1	first
A cop plan on two trees	P2		first
Triangle against square	P3		first
Escape on five- and six-dot rings	P4		first
Fewest added roads so C wins		P2	return
Design a cop-win and a robber-win map		P3	return
Fold F1 and F2 down		P4	return
Two cops from a on the six-ring		P5	return
Two cops on the eight-ring and on every ring		P6	return

First route. Grades 2–5 P1–P4; Grades 4–5 P1. **Return visits.** Grades 2–5 none; Grades 4–5 P2–P6.

Key, one entry per problem

Best cop start on two paths

Grades 2–5 P1 · Grades 4–5 P1 · first route

Idea. On a path the middle dot is the best start: C walks toward R, and R on a farthest end forces about half the path's length in cop turns.

On the five-dot path start C at c; against the best escape choices capture takes at most two cop moves. On the seven-dot path start at d; at most three suffice. R at a farthest endpoint can stay, forcing the same lower bound. Any other start has a farther endpoint, so it cannot improve the worst case. C follows the unique path to R; R may move or stay but cannot cross C or pass the endpoint. Counting a round as two player turns instead of one cop move changes the reported number, so identify the quantity when discussing speed.

A cop plan on two trees

Grades 2–5 P2 · first route

Idea. Every tree is cop-win: with C's dot removed, the piece of the tree holding R shrinks with every cop step toward R.

Both maps are trees. C can start anywhere and repeatedly take the first edge of the unique path toward R. Starting at c in either supplied map catches within two cop moves, but the printed task asks for a guaranteed strategy, not the fastest count.

For the general reason, delete C's dot and shade the connected component containing R. C's next move goes into that component. Unless it captures R, deleting the new C dot leaves R in a strict subset of the former component. R's reply cannot leave this subset without crossing C. Repeated strict shrinking in a finite graph forces capture. Merely saying the distance decreases is insufficient: R can move away and restore the distance for a while. The territory cut off by C, not the current distance alone, is what accumulates.

Triangle against square

Grades 2–5 P3 · first route

Idea. One cop catches on the triangle at once; on the square R stays opposite C forever.

On the triangle C can start anywhere and capture the distinct starting R on its first move. On the square R starts opposite C. Whenever C stays, R stays; whenever C moves one edge, R moves one edge to the new opposite dot. Before every cop turn there are two roads between them, so the next cop move cannot capture. Every possible cop reply is accounted for; this explains perpetual escape, unlike a long lucky game.

Escape on five- and six-dot rings

Grades 2–5 P4 · first route

Idea. On a bare ring of four or more dots R escapes forever by staying outside C's one-road reach.

On both the five- and six-dot ring, R begins at a nonneighbor of C. Before each cop turn, keep R outside C's closed one-road neighborhood. After C moves, stay if still safe; otherwise C is now a neighbor, so R steps away along the other ring edge. For a bare ring of at least four vertices that neighbor is neither C nor adjacent to C, restoring safety. "Opposite" is not an adequate general rule on an odd ring. Ask a partner to challenge the suggested response by staying or choosing either direction.

Fewest added roads so C wins

Grades 4–5 P2 · return route

Idea. One diagonal makes the square cop-win; the pentagon needs two chords, and one chord still leaves R a square to run on.

The square needs exactly one diagonal. With none, the previous escape strategy works; either diagonal makes its endpoint adjacent to every dot, so C starts there and catches on move one.

The pentagon needs two chords. Chords a–c and a–d make a adjacent to every other dot, so two suffice. Zero fails by the ring strategy. To show one fails, any one-chord pentagon can be relabeled with chord a–c. Keep R on the square a–c–d–e–a. Project a cop at b to a, leaving other cop positions unchanged. Every actual cop move projects to a legal square move or stay, since b's neighbors a,c both map next to a. R uses its square escape strategy against that projected cop. Actual capture would imply projected capture, which never occurs. This is a graph-map proof, not just the report of several games. Children may test all five single-chord choices together before an adult offers this explanation.

A different pair of chords is acceptable if its claimed strategy is valid; the question asks a minimum, not every minimum pair. Adding roads can help a cop by removing hiding places, or help an escapee by creating a way around a former barrier. The number of roads alone does not decide the outcome.

Design a cop-win and a robber-win map

Grades 4–5 P3 · return route

Idea. Six dots and six roads force a cycle, yet such maps can go either way: a triangle with a tail is cop-win, the six-ring robber-win.

One cop-win example uses the triangle a–b–c–a with tail c–d–e–f: six dots and six roads. Start C at c. Catch an initial robber at a or b immediately; otherwise sweep the tail. R cannot pass through C to use the triangle. The bare six-cycle is a robber-win example with the same dot/road counts. These examples show that a cycle's presence does not by itself imply escape.

The minimum six-road condition makes a tree unavailable as the cop example, so it serves this comparison rather than an arbitrary drawing restriction. Accept other connected maps with valid response strategies; do not judge from visual resemblance. A held hint is to attach a branch to a board whose behavior the children already understand. Crossings drawn without a dot do not create roads between the crossing routes.

Fold F1 and F2 down

Grades 4–5 P4 · return route

Idea. A map is cop-win exactly when folds can shrink it to one dot; F1 folds to a dot, F2 sticks at a square.

Materials p.1 supplies F1 and its working copy; p.2 supplies F2 and its working copy. Keep the original visible, but test each fold in the current reduced graph. Staying counts among reachable places. Deleting a dot deletes its incident roads; do not create a replacement road through it.

F1 can fold e into d, then d into c, then f into a. The remaining triangle a,b,c folds a into b, then b into c. Each inclusion can be checked from the actual remaining board. Other legal orders work. Its one-dot reduction certifies that one cop has a strategy.

In F2 only b is initially removable: it can fold into a or c. Either removal leaves the bare square on a,c,d,e. No adjacent pair in a square has one closed neighborhood contained in the other, so no further fold is possible. F2 is robber-win by the projected-square strategy above. A single dead end in an arbitrary search would need justification; here both initial choices produce the same residual square. Children can stop at a checkable fold certificate; understanding the simultaneous real/shadow strategy is an additional readiness gate.

Two cops from a on the six-ring

Grades 4–5 P5 · return route

Idea. Two cops catch on a ring: one holds a dot as a blocker while the other sweeps the path that remains.

For the six-dot task both cops enter at a; the same start works for the eight-dot and general-ring question. Keep C1 at a as a blocker. On its first turn C2 moves to the adjacent b, then advances along the remaining path toward R. Deleting a turns the other ring vertices into that path. R cannot pass C1 to get around the far end, nor pass C2, so the path strategy captures. The same plan works on every finite cycle. If R chooses the occupied starting dot, capture is

immediate; if it starts at b, C2 catches on its first move. Two cops are necessary on a bare cycle of length at least four because the one-cop escape invariant applies; the triangle only needs one.

Two cops on the eight-ring and on every ring

Grades 4–5 P6 · return route

Idea. The blocker-and-sweeper plan works on every ring, whatever its number of dots.

Both cops may each take at most one road or stay, then R gets one turn. Two moves are not a pooled budget that one cop may spend. The six-dot example gives a concrete coordination attempt before the eight-dot/general question. A held hint is to ask whether one cop could make one part of the ring permanently unusable. A strategy in which both cops move can also be valid; require an explanation of the region R can still occupy.